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From Chaos to Order: Organizing Dhaka's Cake Sellers Using Algorithms

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Abstract

This report details the implementation of a computational solution to address the unorganized operation of cake sellers in Dhaka City, which contributes to urban crises like traffic and hygiene issues. The project was completed through three distinct milestones utilizing fundamental data structures and algorithms. Milestone 1 focused on data management via the comparison of five sorting algorithms and the implementation of binary search. Milestone 2 addressed the spatial organization problem by allocating 100 cake sellers to 15 designated road network nodes based on proximity and fixed capacity. Milestone 3 implemented a shortest path algorithm to find efficient routes between locations within the city network. The comprehensive solution successfully demonstrates the application of efficient algorithms for urban resource management and spatial optimization.

Introduction

Dhaka City faces significant urban challenges, exacerbated by the informal operation of an estimated 2 million street food vendors. The project specifically addresses a localized problem in the Agargaon area concerning the random distribution of cake sellers, which creates traffic congestion, hygiene, and administrative difficulties. The core objective of this project was to leverage algorithmic knowledge to impose a systematic order on this chaotic system.

The developed system manages a dataset of 100 cake sellers and utilizes a model of the city's road network composed of 15 interconnected nodes. This report presents the design, implementation, and results of the proposed algorithmic solutions for sorting, searching, spatial allocation, and route optimization.

Problem Statement and Proposed Solution

The primary problem is the unsystematic, capacity-unaware distribution of 100 cake sellers across a defined city area. The proposed solution is a multi-phased algorithmic framework designed to: (1) efficiently manage and query seller data, (2) spatially allocate sellers to fixed city nodes, and (3) provide optimal navigation for city personnel.

Milestone 1: Sorting & Searching Cake Sellers

1.Sorting Algorithms Comparison: The initial step involved sorting the 100 cake sellers based on their Hygiene Score (40-100). Five sorting algorithms were implemented and compared: Insertion, Merge, Quick, Bubble, and Counting Sort.

Counting Sort was found to be the best algorithm for sorting the hygiene scores. Unlike the comparison-based algorithms (Merge, Quick, Insertion, Bubble), which have a best-case time complexity $\Omega(n \log n)$ or $\Omega(n)$, Counting Sort is a non-comparison algorithm with a time complexity of $O(n+k)$, where 'n' is the number of elements and 'k' is the range of input values. Since the hygiene score is strictly bounded from 40 to 100 resulting in a small ranged(k) value, its performance is drastically superior, demonstrating the fastest execution time and the lowest total steps.

2.Searching Algorithm Implementation: To quickly retrieve the detailed information of a specific cake seller, a Binary Search algorithm was implemented on a list of sellers sorted by their unique ID. Binary Search, with a worst-case time complexity of $O(\log n)$, is highly efficient for large datasets. In an example search for CAKE-005 out of 100 sellers, the algorithm successfully located the seller in only 7 steps, confirming its practical efficiency. However, we can also use Linear search directly without sorting the dataset as its worst case is $O(n)$. But for large dataset Linear search isn't efficient. So, we prefer to sort the data first and then implement Binary search.

Milestone 2: Seller Allocation

The core problem of organization was addressed by allocating the 100 cake sellers to the 15 city nodes, where each node has a fixed capacity of 5 slots.

Proposed Solution: A greedy, distance-based allocation approach was used. For each seller, the Euclidean distance to all 15 nodes was calculated, and the seller was assigned to the nearest node that had available capacity.

Results: The total available capacity across all nodes was $(15 \times 5) = 75$ slots.

- Total Assigned Sellers: 75.
- Total Waitlisted Sellers: 25 (100 total sellers - 75 assigned sellers).

The results show a full utilization of the available infrastructure, with all 15 nodes having 0 remaining slots. The remaining 25 sellers were placed on a waitlist, creating a clear administrative mechanism for managing the surplus vendor population.

Milestone 3: Shortest Pathfinding

Following the allocation, the final milestone involved implementing a robust pathfinding mechanism to aid in navigation, such as for regulatory visits or customer deliveries.

Proposed Solution: Given the presence of a defined 15-node city road network, a Shortest Path Algorithm Dijkstra's was applied to the graph representation of the road network. This algorithm calculates the path with the minimum total distance between a starting node (the inspector's location) and a destination node (the allocated seller's node).

Results: A successful route calculation was demonstrated. For a scenario starting at Node 2 and visiting the seller at Node 5, the shortest route found was: $2 \rightarrow 1 \rightarrow 4 \rightarrow 5$. The output provided the number of stops (4 nodes) and the corresponding total distance of the route, confirming the functionality of the implemented pathfinding algorithm.

Limitations

1. Allocation Optimality: The allocation in Milestone 2 used a greedy, nearest-neighbor approach which prioritizes local optimality (each seller going to their single closest available node). This does not guarantee a globally optimal solution that minimizes the *sum* of distances for all 75 assigned sellers, potentially leading to a suboptimal overall configuration.
2. Counting Sort Range: The superior performance of Counting Sort is contingent on the small, known range of the Hygiene Score (40-100). This algorithm cannot be efficiently extended to sort other seller attributes with large, continuous, or unbounded ranges, such as monthly revenue or GPS coordinates.
3. Pathfinding Granularity: The pathfinding algorithm operates solely on the 15 abstract nodes of the city network. In reality, a cake seller's exact location is a GPS coordinate, and the current system approximates this location to the nearest node. This abstraction limits the real-world precision and accuracy of the shortest route calculation for the "last mile" travel.
4. Fixed Capacity: The capacity of 5 sellers per node is static. It does not account for dynamic variables such as demand, peak hours, or seller revenue, which would ideally inform a more adaptive, weighted capacity model.

Conclusion and Future work

The project successfully applied various fundamental algorithms to model and solve a real-world urban management challenge concerning the organization of cake sellers in Dhaka. The system provides efficient data handling (Counting Sort for fast grading, Binary Search for quick retrieval), a structured allocation of vendors to fixed points, and practical route optimization (Shortest Path Algorithm) for navigation.

Future work should focus on:

1. Optimal Allocation: Re-implementing the allocation logic using a more advanced optimization algorithm, such as Max Flow/Min Cost Flow or a sophisticated clustering technique, to ensure a globally optimal assignment that minimizes the total aggregate travel distance for all assigned sellers.
2. Dynamic Capacity Modeling: Integrating the seller's monthly revenue and experience into the allocation model to create a dynamic capacity per node. This would allow better-performing sellers or high-demand areas to occupy more space.
3. Advanced Pathfinding: Upgrading the Shortest Path Algorithm to an informed search technique like the A* algorithm, using a geographical heuristic (e.g., Euclidean distance) to significantly reduce the search space and improve efficiency on a larger, more realistic city road network graph.
4. Scalability: Expanding the underlying road network graph from 15 nodes to a larger, data-driven representation of the actual city's road network to ensure the system's viability for city-wide implementation.

References

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4. Shortest Path: [Dijkstra's Algorithm - GeeksforGeeks](#)